

Let  $K \subset V$  be a convex set. A **face** of  $K$  is a nonempty convex subset  $F \subset K$  such that if  $x \in F$  and  $x$  can be written as a positive convex combination of  $x_1, \dots, x_m \in K$ , then  $x_1, \dots, x_m \in F$ .

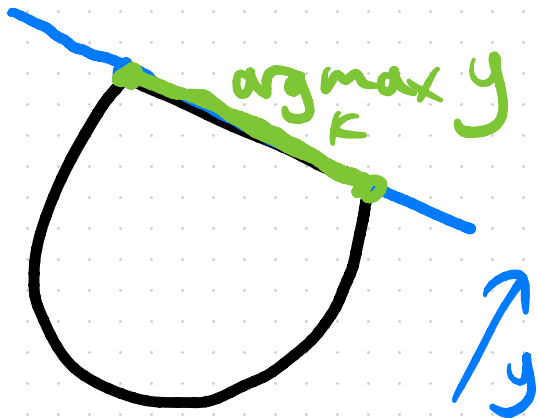
For cones, "convex combination" can be replaced by "conical combination".

If  $F$  is a face of  $K$  and  $G \subset F$ , then  $G$  is a face of  $F$  iff it is a face of  $K$ .

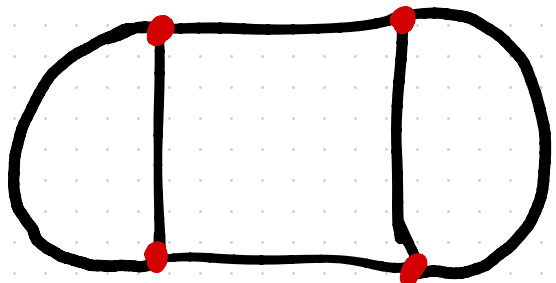
A support set of a convex set  $K$  is a nonempty set of the form  $\operatorname{argmax}_K y$ , where  $y \in V^*$ .

set of all  $x \in K$   
such that  $y(x) \geq y(x')$   
for all  $x' \in K$ .

Every support set of  $K$  is a face of  $K$ .



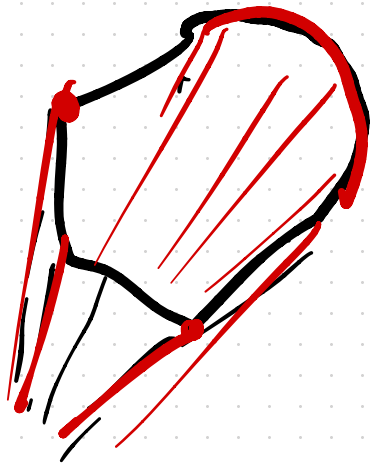
If  $y \geq 0$  and  $K \neq \emptyset$   
 $\operatorname{argmax}_K y = K$



↑  
faces, but not  
support sets

An **extreme ray** of a convex cone is a face of the cone which is a ray.

Thm: Every closed convex cone with lineality space  $\{0\}$  is the conical hull of its extreme rays.



Proof: We induct on the dimension of  $V$ , the ambient space.

True for  $\dim V = 1$ .

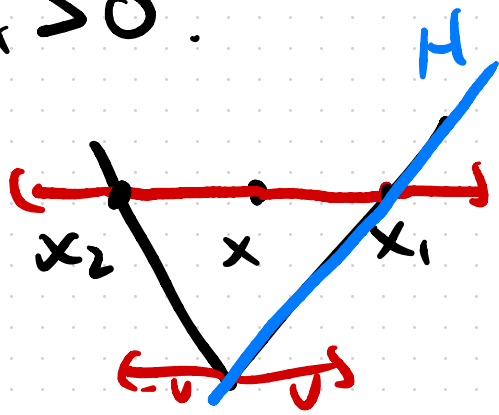
Assume  $\dim V \geq 2$ .

Let  $C$  be a closed convex cone with  $\text{lin } C = \{0\}$ .

Let  $x \in C$ . (Since  $C$  is closed and  $\text{lin } C = \{0\}$ ,

There exist some  $v \in V$  such that  $v, -v \notin C$ .)  
We'll prove later.

Thus  $x + \lambda v, x - \lambda v \notin C$  for large enough  $\lambda > 0$ .



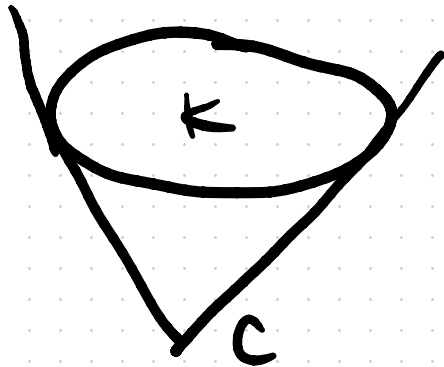
In particular, there is some  $x_1 = x + \lambda_1 v$  on the boundary of  $C$  for  $\lambda_1 \geq 0$ .

By the supporting hyperplane theorem, there is a hyperplane  $H$  supporting  $C$  at  $x_1$ , with  $H$  passing through  $O$ . Then  $H \cap C$  is a cone in the subspace  $H$  with lineality space  $\{0\}$ . Therefore,  $x_1$  can be written as a conical combination of the extreme rays of  $C \cap H$ . There are also extreme rays of  $C$ , so  $x_1$  is a conical combination of extreme rays of  $C$ .

Same for  $x_2 = x - \lambda_2 v$ . Thus  $x$  can be written as a conical combination of extreme rays of  $C$ .

An **extreme point** is a line which is a point.

Corollary: A bounded closed convex set is the convex hull of its extreme points.



$$\begin{aligned} K &\subset V \\ C &\subset V \times \mathbb{R} \\ &\text{"} \\ &\text{cone}(K \times \{1\}) \end{aligned}$$

Theorem: A polyhedral cone has finitely many extreme rays.

Proof: A polyhedral cone can be written as

$$C = \{x \in V : y_i(x) \leq 0, i \in I\}$$

where  $\{y_i\}_{i \in I}$  is a finite subset of  $V^*$ .

Let  $r$  be an extreme ray of  $C$ .

Let  $H_i$  be the hyperplane  $\{y_i(x) = 0\}$



We claim  $r$  must be in one of the  $H_i$ .

If not, then  $y_i(x) < 0$  for some  $x \in r$  and all  $i \in I$ . Then  $x$  is in the interior of  $C$ .

Then  $x$  can be written as the conical hull of vectors not in  $r$ , so  $r$  is not a face.  $\Rightarrow \Leftarrow$

So  $r$  is in  $H_i$  for some  $i$ .  $H_i \cap C$  is a polyhedral cone, and a <sup>(support set)</sup> face of  $C$ . Then  $r$  is an extreme ray of  $H_i \cap C$ .

By induction of the ambient dimension,  $H_i \cap C$  has

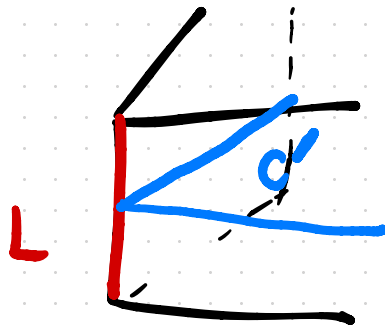
finitely many extreme rays.

Thus every extreme ray of  $C$  is an extreme ray of  $H_i \cap C$  for some  $i$ , each of which has finitely many extreme rays, and there are finitely many  $H_i \cap C$ .

Corollary: A polyhedral cone is the conical hull of a finite set of vectors.

Proof: If  $\text{lin } C = \{0\}$ , then this follows from the previous two theorems.

If  $\text{lin } C \neq \{0\}$ , let  $L = \text{lin } C$ . We can write  $C = L \oplus C'$  where  $C'$  is a polyhedral cone with  $\text{lin } C' = \{0\}$ .



$$C = C' \oplus L$$

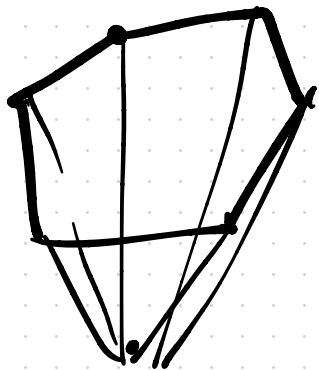
$C'$  is the conical hull of a finitely many extreme rays.  
 $L$  is a vector space, so it can be written as the conical hull of a finite set of vectors.



Thus  $C$  is the conical hull of a finite set of vectors.

A **polytope** is a bounded polyhedron.

Carollary: A polytope has finitely many extreme points (**vertices**), and is the convex hull of its vertices.



Theorem: The conical hull of a finite set of vectors is a polyhedral cone.

Corollary: The convex hull of a finite set of points is a polytope.